


# BANGLADESH UNIVERSITY OF PROFESSIONALS

## DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

# Digital Communication

Digital Modulation

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- Digital Modulation.
- Binary Modulation.
- M-ary Modulation.

## ❑ Digital Modulation

Digital modulation is the process of converting ***digital data (bits such as 0s and 1s)*** into *Analog signals* by ***changing certain properties*** of a ***high-frequency carrier signal*** so that the data can be ***transmitted*** through a ***communication channel***.

The carrier signal properties/characteristics that can be changed are:

- Amplitude.
- Frequency.
- Phase.

## ❑ Digital Modulation Necessities

### 1. Long-Distance Transmission

High-frequency modulated signals can travel much farther than low-frequency digital signals.

### 2. Efficient antenna size

Antenna height is inversely related to frequency. Higher carrier frequencies allow practical antenna sizes.

$$L \propto \lambda,$$

where, $\lambda$ = wavelength of a transmitted signal

$L$ = the height of the antenna

## ❑ Digital Modulation Necessities

### 2. Efficient antenna size

Antenna height is inversely related to frequency. Higher carrier frequencies allow practical antenna sizes.

$$\boldsymbol{L \propto \lambda} \ldots\ldots\ldots\ldots (1)$$

where, $\boldsymbol{\lambda}$ = wavelength of a transmitted signal

$\boldsymbol{L}$ = the height of the antenna

Let's say, the constant is $\frac{1}{4}$. Then,

$$L = \frac{\boldsymbol{\lambda}}{\boldsymbol{4}} \ldots\ldots\ldots\ldots (2)$$

- **Digital Modulation Necessities**

**2. Efficient antenna size**

$$L = \frac{\boldsymbol{\lambda}}{\mathbf{4}} \ldots\ldots\ldots\ldots (2)$$

The relationship between wavelength, $\boldsymbol{\lambda}$, and frequency, $\boldsymbol{f}$, is reciprocal, and formally,

$$\boldsymbol{\lambda} = \frac{\boldsymbol{c}}{\boldsymbol{f}} \ldots\ldots\ldots\ldots (3)$$

where, $\boldsymbol{\lambda}$ = wavelength of a transmitted signal

$\boldsymbol{f}$ = the height of the antenna

$\boldsymbol{c}$ = speed of light, $3 \times 10^8$ m/s

- **Digital Modulation Necessities**

**2. Efficient antenna size**

$$L = \frac{\boldsymbol{\lambda}}{\mathbf{4}} \ldots\ldots\ldots\ldots (2)$$

The relationship between wavelength, $\boldsymbol{\lambda}$, and frequency, $\boldsymbol{f}$, is reciprocal, and formally,

$$\boldsymbol{\lambda} = \frac{\boldsymbol{c}}{\boldsymbol{f}} \ldots\ldots\ldots\ldots (3)$$

Then, from (2), and (3),

$$\boldsymbol{L} = \frac{\boldsymbol{c}}{\mathbf{4}\boldsymbol{f}} \ldots\ldots\ldots\ldots (4)$$

Therefore, the antenna length will decrease if the frequency of the signal is increased.

## ❑ Digital Modulation Necessities

3. **Noise immunity**

Digital modulation provides better resistance to noise and interference compared to Analog communication.

- Again, the receiver needs to recognize/identify in stead of reconstruct/recover.

4. **Multiplexing capability**

Multiple signals can be transmitted simultaneously using different carrier frequencies.

- Digital modulation makes multiplexing easier and more efficient because different modulated signals can be separated using frequency, time, or code.

## ❑ Digital Modulation Necessities

5. **Efficient bandwidth usage**

   Modern digital modulation techniques use bandwidth more efficiently.

   - Advanced digital modulation techniques can send multiple bits in a single signal symbol.
   - This increases the data rate without proportionally increasing bandwidth.

6. **Compatibility with wireless communication**

   Radio, satellite, Wi-Fi, Bluetooth, and cellular systems require modulated carrier waves.

   - Wireless systems require the data to be placed onto a high-frequency carrier wave using modulation.

# Digital Binary Modulation

## Digital Binary Modulation

- Digital modulation technique in which binary data (0 and 1) is transmitted by changing one parameter of a carrier signal.
- Since the data contains only two binary states:0 and 1,the modulation uses two distinct signal conditions to represent them.
- Each bit is represented by one complete signal waveform (called a symbol).
- The mapping/how the symbol would look like depends on the type of the digital binary modulation
    1. Binary Amplitude Shift Keying (BASK / ASK)
    2. Binary Frequency Shift Keying (BFSK / FSK)
    3. Binary Phase Shift Keying (BPSK)

# Digital Binary Modulation

## ❑ Binary Amplitude Shift Keying (BASK / ASK)

- In BASK, the amplitude of the carrier changes according to the binary data.
  - Bit 1 → carrier transmitted
  - Bit 0 → carrier absent or lower amplitude

- Let's say the carrier signal, $\boldsymbol{c(t) = A\cos(2\pi f_c t + \phi)}$

where, $A$ = Amplitude of the signal

$t$ = time variables (in seconds)

$\cos$ = Cosine waveform function

$f_c$ = Carrier frequency (Hz)

$2\pi f_c t$ = Angular variation of the carrier with time

$\phi$ = Phase angle (in radians or degrees)

# Digital Binary Modulation

## ❑ Binary Amplitude Shift Keying (BASK / ASK)

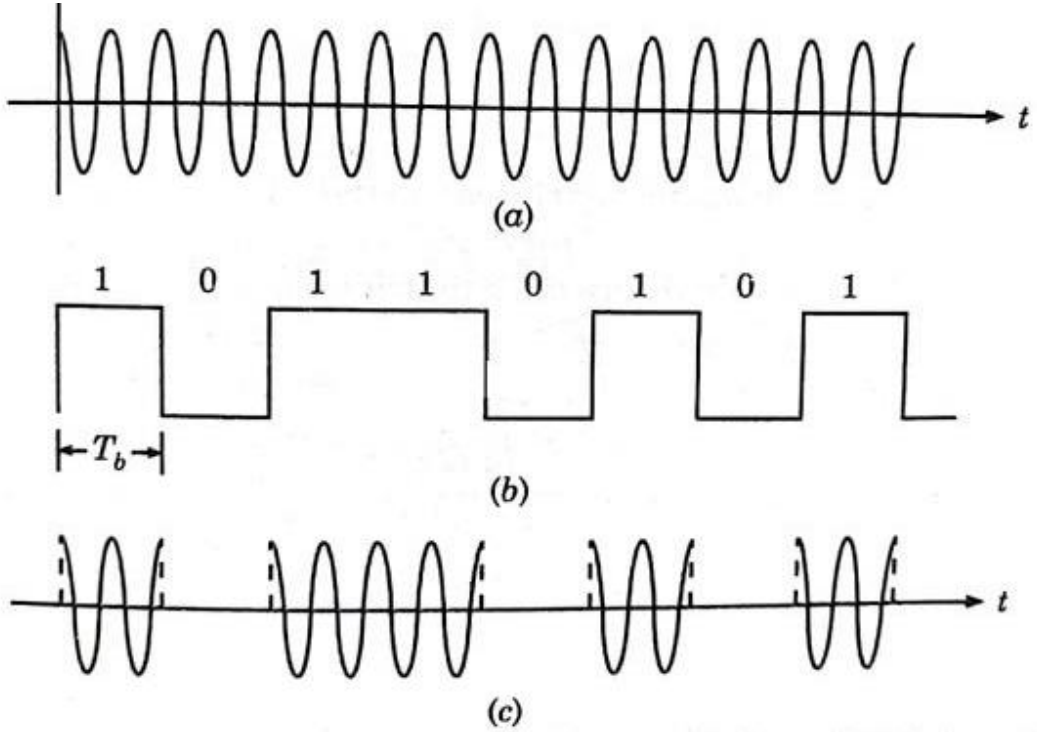
- Mathematical Representation of BASK/ASK:
  - $$s(t) = \begin{cases} A\cos(2\pi f_c t), & for\ bit\ 1 \\ 0, & for\ bit\ 0 \end{cases}$$

    where, $s(t)$ = transmitted/modulated signal

  - Only the amplitude changes according to the bit.
  - Frequency and phase remain fixed.
  - $\phi$ is truncated.
  - Simplest among the Digital Modulation techniques.
  - Simple multiplication of the bit values and carrier signal.

# Digital Binary Modulation

- **Binary Amplitude Shift Keying (BASK / ASK)**



# Digital Binary Modulation

- **Binary Amplitude Shift Keying (BASK / ASK)**
  - **Pros**
    - Simple and inexpensive implementation.
    - Easy transmitter and receiver design.
    - Requires relatively low bandwidth.
  - **Cons**
    - Highly sensitive to noise and interference
    - Poor power efficiency
    - Less reliable for long-distance communication
  - **Applications**
    - Optical fiber communication
    - RFID systems, and Infrared remote controls

## ❑ Binary Frequency Shift Keying (BFSK / FSK)

- In BFSK, different frequencies represent binary values.
  - Bit 1 → carrier frequency, $f_1$
  - Bit 0 → carrier frequency, $f_0$
- Let's say the carrier signal, $\boldsymbol{c(t) = Acos(2\pi f_c t + \phi)}$

where, $A$ = Amplitude of the signal

$t$ = time variables (in seconds)

cos = Cosine waveform function

$f_c$ = Carrier frequency (Hz)

$2\pi f_c t$ = Angular variation of the carrier with time

$\phi$ = Phase angle (in radians or degrees)

# Digital Binary Modulation

## ❑ Binary Frequency Shift Keying (BFSK / FSK)

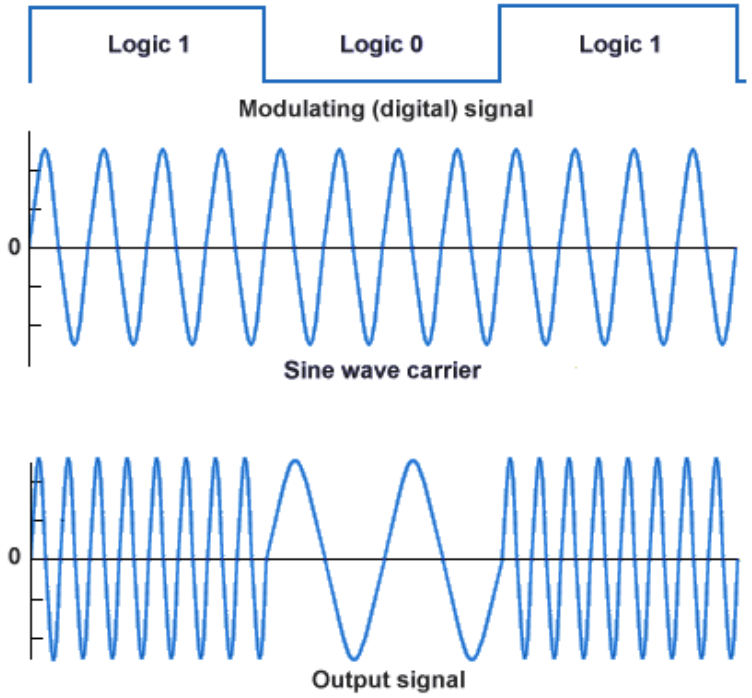
- Mathematical Representation of BASK/ASK:
  - $s(t) = \begin{cases} A\cos(2\pi f_1 t), & for\ bit\ 1 \\ A\cos(2\pi f_0 t), & for\ bit\ 0 \end{cases}$

    where, $s(t)$ = transmitted/modulated signal

  - Only the frequency changes according to the bit.
  - Amplitude and phase remain fixed.
  - $\phi$ is truncated.

# Digital Binary Modulation

- **Binary Frequency Shift Keying (BFSK / FSK)**



# Digital Binary Modulation

## ❑ Binary Frequency Shift Keying (BFSK / FSK)

- **Pros**
  - Better noise immunity than BASK
  - Constant amplitude reduces effects of signal fading
  - More reliable transmission
- **Cons**
  - Requires larger bandwidth
  - More complex circuitry than BASK
  - Lower bandwidth efficiency
- **Applications**
  - Modems
  - Radio telemetry
  - Caller ID systems,
  - Low-speed data communication

## ❑ Binary Phase Shift Keying (BPSK / PSK)

- In BASK, the phase of the carrier changes according to the binary data.
  - Bit 1 → carrier phase $0^0$
  - Bit 0 → carrier phase $180^0$

- Let's say the carrier signal, $\boldsymbol{c(t) = Acos(2\pi f_c t + \phi)}$

where, $A$ = Amplitude of the signal

$t$ = time variables (in seconds)

cos = Cosine waveform function

$f_c$ = Carrier frequency (Hz)

$2\pi f_c t$ = Angular variation of the carrier with time

$\phi$ = Phase angle (in radians or degrees)

## ❑ Binary Phase Shift Keying (BFSK / FSK)

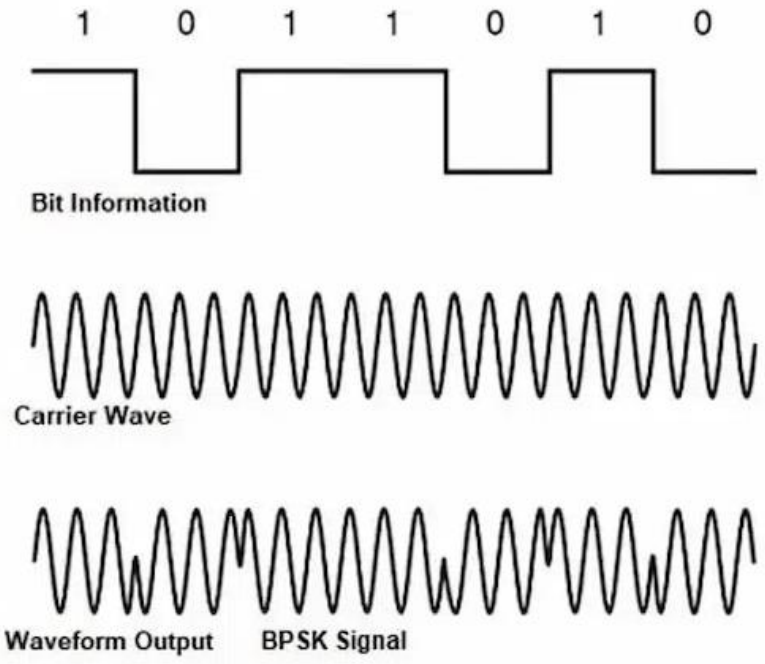
- Mathematical Representation of BASK/ASK:
  - $s(t) = \begin{cases} Acos(2\pi f_c t), & for\ bit\ 1 \\ -Acos(2\pi f_c t), & for\ bit\ 0 \end{cases}$

    where, $s(t)$ = transmitted/modulated signal

  - $\cos(\theta + 180^0) = -\cos(\theta)$
  - Only the phase changes according to the bit.
  - Amplitude and frequency remain fixed.
  - values of $\phi$ is applied.

# Digital Binary Modulation

- **Binary Frequency Shift Keying (BASK / ASK)**



# Digital Binary Modulation

## ❑ Binary Phase Shift Keying (BPSK / PSK)

- **Pros**
  - Excellent noise immunity
  - Better error performance
  - Power efficient
  - Reliable for long-distance communication
- **Cons**
  - More complex receiver design
  - Requires phase synchronization at the receiver
  - Lower data rate compared to higher-order PSK/QAM systems
- **Applications**
  - Wi-Fi systems
  - Satellite communication
  - Deep-space communication
  - Mobile communication systems

## ❑ Comparison Among the Digital Binary Modulation Techniques

| Feature | BASK | BFSK | BPSK |
|---|---|---|---|
| Changed Parameter | Amplitude | Frequency | Phase |
| Noise Immunity | Low | Medium | High |
| Bandwidth Requirement | Low | High | Medium |
| Complexity | Simple | Moderate | Higher |
| Power Efficiency | Low | Medium | High |
| Reliability | Low | Medium | High |